

Miscellaneous Theorems and Ideas

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This is just a collection of some cool facts and theorems I have found, along with their proofs.

Theorem 1.

For an odd positive integer n , and positive integers a, b with $a^2 \leq b < (a+1)^2$ we have the following,

$$\left\lfloor (a + \sqrt{b})^n \right\rfloor = (a + \sqrt{b})^n + (a - \sqrt{b})^n.$$

Proof of Theorem 1.

Firstly, due to Binomial expansion,

$$\begin{aligned} (a + \sqrt{b})^n + (a - \sqrt{b})^n &= \sum_{r=0}^n \binom{n}{r} (\sqrt{b})^r a^{n-r} + \sum_{r=0}^n \binom{n}{r} (-\sqrt{b})^r a^{n-r} \\ &= \sum_{r=0}^n \binom{n}{r} (\sqrt{b})^r [1 + (-1)^r] a^{n-r} \\ &= 2 \sum_{\substack{r=0 \\ r \text{ even}}}^n \binom{n}{r} (\sqrt{b})^r a^{n-r} \end{aligned}$$

which is clearly an integer. Now it suffices to show that

$$(a + \sqrt{b})^n - 1 < (a + \sqrt{b})^n + (a - \sqrt{b})^n \leq (a + \sqrt{b})^n$$

That is

$$-1 < (a - \sqrt{b})^n \leq 0$$

but we know that $a \leq \sqrt{b} < a+1$ and so $a - \sqrt{b} \leq 0$ and $-1 < a - \sqrt{b}$ and raising both inequalities to the power of n gives the desired result. ■